Drift Diffusion

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1 Drift-Diffusion

The continuum flux of a point defect is

$$j_i = -D_{ij}\left(c_{,j} + \frac{c}{kT}E^e_{,j}\right) \tag{1}$$

where E^e is the interaction energy between the defect and the strain field at the equilibrium position. Following, if we let $D_{ij} = \tilde{D}_{ij}\exp\left(\frac{E^e}{k_BT}\right)$ then we obtain:

$$j_i = -\tilde{D}_{ij}\left(c_{,j}\exp\left(\frac{E^e}{k_BT}\right) + \frac{c}{kT}\exp\left(\frac{E^e}{k_BT}\right)E^e_{,j}\right) = -\tilde{D}_{ij}\left(c\exp\left(\frac{E^e}{k_BT}\right)\right)_{,j} = -\tilde{D}_{ij}\tilde{c}_{,j} \tag{2}$$

where we have defined the renormalized concentration

$$\tilde{c}(\boldsymbol{x}) = c(\boldsymbol{x})\exp\left(\frac{E^e(\boldsymbol{x})}{k_BT}\right) \tag{3}$$

The renormalized diffusion tensor can be derived from lattice theory [1, 2]. It is

$$\tilde{D}_{ij}(\boldsymbol{x}) = \frac{1}{4}D_0\sum_{\boldsymbol{h}}\hat{h}_i\hat{h}_j\exp\left(-\frac{E^{s,\boldsymbol{h}}(\boldsymbol{x})}{k_BT}\right) \tag{4}$$

where $E^{s,\boldsymbol{h}}(\boldsymbol{x})$ is the saddle-point interaction energy for a point defect initially located at $\boldsymbol{x}$ and performing a jump $\boldsymbol{h}$. Now consider the function E:

$$E^{s,\boldsymbol{h}}(\boldsymbol{x}) = -P^{s,\boldsymbol{h}}_{ij}\varepsilon_{ij}(\boldsymbol{x}) = -P^{s,\boldsymbol{h}}_{ij}\left(\varepsilon^a_{ij}(\boldsymbol{x}) + \varepsilon^d_{ij}(\boldsymbol{x})\right) \tag{5}$$

where $\boldsymbol{P}^{s,\boldsymbol{h}}$ is the elastic dipole tensor of the defect at the saddle point of the jump trajectory $\boldsymbol{h}$, and the elastic strain is decomposed in applied plus dislocation. The dipoe tensor is considered constant, that is its polarizability due to the srain field is neglected.

First-order expansion of (4) yields

$$\tilde{D}_{ij}(\boldsymbol{x}) = D_0\delta_{ij} + \underbrace{\frac{1}{4}\frac{D_0}{kT}\sum_{\boldsymbol{h}}\hat{h}_i\hat{h}_jP^{s,\boldsymbol{h}}_{kl}\varepsilon^a_{kl}(\boldsymbol{x})}_{\text{SIPA-AD-Woo}} + \underbrace{\frac{1}{4}\frac{D_0}{kT}\sum_{\boldsymbol{h}}\hat{h}_i\hat{h}_jP^{s,\boldsymbol{h}}_{kl}\varepsilon^d_{kl}(\boldsymbol{x})}_{\text{EID}} \tag{6}$$

Diffusion far from dislocations Away from dislocations, and under a quasi uniform strain field, the renormalized diffusion tensor becomes a constant and generally anisotropic offset of the original diffusion coefficient

$$\tilde{D}_{ij} = D_0\delta_{ij} + \frac{1}{4}\frac{D_0}{kT}\sum_{\boldsymbol{h}}\hat{h}_i\hat{h}_jP^{s,\boldsymbol{h}}_{kl}\varepsilon^a_{kl} \tag{7}$$

In this case, also the drift term in the original diffusion equation drops out, and the drift-diffusion problem reduces to a standard diffusoin problem with constant but stress-dependent diffusion coefficient. Anisotropic diffusion affects both diffusinal and dislocation creep. The effect on dislocation creep is due to the the fact that line elements receive a net flux that is different depending on their orientation in the anisotropic diffusion field. This is the Stress Induced Preferential Absorpition due to Anosotpropic Diffusion (SIPA-AD) proposed by Woo. Because the change in diffusion coefficient is proportional to the applied elastic strain, the stress exponent of SIPA-AD-Woo is expected to be 1.

SIPA-AD-Woo is a first order-effect in the defect dipole tensor. If the polarizability due to the strain field is considered, other second-order SIPA effects emerge (SIPA-I and SIPA-AD-Dederichs).

Diffusion near dislocations

2 Modification of the diffusion–dislocation coupling

For each mobile defect species s, let $\boldsymbol{j}_s$ denote the defect-number flux, while p_s, G_s, and R_s are rates of change of site fraction. The steady balance law is therefore written as

$$-\Omega_0 \boldsymbol{\nabla} \cdot \boldsymbol{j}_s + p_s + G_s + R_s(\boldsymbol{c}) = 0, \tag{8}$$

where p_s is the line-supported production term associated with dislocation climb, G_s is the irradiation production rate, and R_s contains the bulk first- and second-order reactions. Introducing

$$H_s(\boldsymbol{x}) = \exp\left(\frac{E_s^e(\boldsymbol{x})}{k_B T}\right), \qquad W_s(\boldsymbol{x}) = H_s(\boldsymbol{x}) c_s(\boldsymbol{x}), \tag{9}$$

transforms the exact drift–diffusion flux into

$$\boldsymbol{j}_s = -\frac{1}{\Omega_0} \widetilde{\boldsymbol{D}}_s(\boldsymbol{x}) \boldsymbol{\nabla} W_s, \qquad \widetilde{\boldsymbol{D}}_s(\boldsymbol{x}) = \widetilde{\boldsymbol{D}}_s\left(\boldsymbol{\sigma}_{\text{ext}} + \boldsymbol{\sigma}_{\text{DD}}(\boldsymbol{x})\right). \tag{10}$$

The equilibrium-position interaction with the total stress is contained in H_s, whereas $\widetilde{\boldsymbol{D}}_s$ describes the stress dependence of the saddle-point migration process. Equation (10) still contains the full diffusion tensor and no approximation has yet been made.

The activity is decomposed as

$$W_s = W_s^{\infty} + \widetilde{W}_s, \tag{11}$$

where W_s^{∞} is the dislocation-associated field and $\widetilde{W}_s$ is a smooth finite-domain correction. The physical total concentration reconstructed from this decomposition remains

$$c_s = H_s^{-1}\left(W_s^{\infty} + \widetilde{W}_s\right). \tag{12}$$

Because Eq. (10) is linear in the activity gradient, this decomposition induces the exact flux decomposition

$$\boldsymbol{j}_s = \boldsymbol{j}_s^{\infty} + \widetilde{\boldsymbol{j}}_s, \tag{13}$$

$$\boldsymbol{j}_s^{\infty} = -\frac{1}{\Omega_0} \widetilde{\boldsymbol{D}}_s(\boldsymbol{x}) \boldsymbol{\nabla} W_s^{\infty}, \qquad \widetilde{\boldsymbol{j}}_s = -\frac{1}{\Omega_0} \widetilde{\boldsymbol{D}}_s(\boldsymbol{x}) \boldsymbol{\nabla} \widetilde{W}_s. \tag{14}$$

Both components therefore contain the same full tensor $\widetilde{\boldsymbol{D}}_s(\boldsymbol{\sigma}_{\text{ext}} + \boldsymbol{\sigma}_{\text{DD}})$ at this stage. We define an exact source partition by assigning the dislocation line source to the analytical component and the smooth bulk terms to the finite-domain component:

$$-\Omega_0 \boldsymbol{\nabla} \cdot \boldsymbol{j}_s^{\infty} + p_s = 0, \tag{15}$$

$$-\Omega_0 \boldsymbol{\nabla} \cdot \widetilde{\boldsymbol{j}}_s + G_s + R_s(\boldsymbol{c}) = 0. \tag{16}$$

The full tensor in Eq. (14) is strongly heterogeneous near a dislocation and does not admit the constant-coefficient anisotropic Green function used by the DDD formulation. For a uniform applied stress, let $\boldsymbol{D}_s^{\mathrm{ext}} = \widetilde{\boldsymbol{D}}_s(\boldsymbol{\sigma}_{\mathrm{ext}})$ and approximate the analytical component flux as

$$\boldsymbol{j}_s^{\infty,\mathrm{app}} = -\frac{Z_s^{\mathrm{EID}}}{\Omega_0}\boldsymbol{D}_s^{\mathrm{ext}}\boldsymbol{\nabla}W_s^{\infty}. \tag{17}$$

Here $Z_s^{\mathrm{EID}}\boldsymbol{D}_s^{\mathrm{ext}}$ represents the residual effect of the dislocation stress on saddle-point transport in the analytical channel; the equilibrium-position interaction is already contained in H_s. Substitution into the line-source balance gives

$$\boldsymbol{\nabla}\cdot\left(\boldsymbol{D}_s^{\mathrm{ext}}\boldsymbol{\nabla}W_s^{\infty}\right) + \frac{p_s}{Z_s^{\mathrm{EID}}} = 0. \tag{18}$$

Thus the physical climb production p_s remains conservative. Let G_s^{ext} denote the Green function with the convention $\boldsymbol{\nabla}\cdot(\boldsymbol{D}_s^{\mathrm{ext}}\boldsymbol{\nabla}G_s^{\mathrm{ext}}) = -\delta$. The analytical solution can then be written as

$$G_{Z,s}^{\mathrm{app}} = \frac{G_s^{\mathrm{ext}}}{Z_s^{\mathrm{EID}}}, \qquad W_s^{\infty} = G_{Z,s}^{\mathrm{app}} * p_s. \tag{19}$$

For the smooth finite-domain component, the dislocation-scale variation of the migration tensor is neglected and its exact flux is approximated as

$$\widetilde{\boldsymbol{j}}_s^{\mathrm{app}} = -\frac{1}{\Omega_0}\boldsymbol{D}_s^{\mathrm{ext}}\boldsymbol{\nabla}\widetilde{W}_s. \tag{20}$$

The exact reaction term in Eq. (16) depends on the total concentration in Eq. (12). To avoid volume quadrature of the sharply peaked analytical field, it is separately approximated by

$$R_s(\boldsymbol{c}) \simeq R_s(\boldsymbol{c}^{\mathrm{bulk}}), \qquad c_s^{\mathrm{bulk}} = H_s^{-1}\widetilde{W}_s. \tag{21}$$

The finite-domain problem is therefore

$$\boldsymbol{\nabla}\cdot\left(\boldsymbol{D}_s^{\mathrm{ext}}\boldsymbol{\nabla}\widetilde{W}_s\right) + G_s + R_s(\boldsymbol{c}^{\mathrm{bulk}}) = 0 \qquad \text{in } \Omega. \tag{22}$$

The two component approximations define the reduced total flux

$$\boldsymbol{j}_s^{\mathrm{red}} = \boldsymbol{j}_s^{\infty,\mathrm{app}} + \widetilde{\boldsymbol{j}}_s^{\mathrm{app}} = -\frac{\boldsymbol{D}_s^{\mathrm{ext}}}{\Omega_0}\left(Z_s^{\mathrm{EID}}\boldsymbol{\nabla}W_s^{\infty} + \boldsymbol{\nabla}\widetilde{W}_s\right), \tag{23}$$

and the closed reduced balance

$$-\Omega_0\boldsymbol{\nabla}\cdot\left(\boldsymbol{j}_s^{\infty,\mathrm{app}} + \widetilde{\boldsymbol{j}}_s^{\mathrm{app}}\right) + p_s + G_s + R_s(\boldsymbol{c}^{\mathrm{bulk}}) = 0. \tag{24}$$

For $Z_s^{\mathrm{EID}} \neq 1$, Eq. (23) is not the single local constitutive law $-\boldsymbol{D}_s^{\mathrm{ext}}\boldsymbol{\nabla}(W_s^{\infty} + \widetilde{W}_s)/\Omega_0$. Instead, it follows from applying different, explicitly stated approximations to the two exact component fluxes in Eq. (14). Neglecting reactions involving W_s^{∞} assumes that diffusion and absorption in the unresolved dislocation-core region are faster than the local bulk reactions. It also ensures that no strongly peaked infinite-domain quantity enters the FEM volume quadrature.

If W_s^{b} denotes the prescribed boundary activity, the Dirichlet condition for the correction is

$$\widetilde{W}_s\Big|_{\partial\Omega} = W_s^{\mathrm{b}} - W_s^{\infty}|_{\partial\Omega}\,. \tag{25}$$

The thermodynamic equilibrium condition on the dislocation line is retained without modification:

$$\left(W_s^{\infty} + \widetilde{W}_s\right)\Big|_{\mathcal{N}} = W_s^{\mathrm{eq}}. \tag{26}$$

Thus the bias changes the climb flux associated with a prescribed activity driving force, but it does not alter the equilibrium activity itself.

For a test function $\delta\widetilde{W}_s$ that vanishes on the Dirichlet boundary, the weak form of Eq. (22) is

$$\int_\Omega \boldsymbol{\nabla}(\delta\widetilde{W}_s)\cdot\left(\boldsymbol{D}_s^{\mathrm{ext}}\boldsymbol{\nabla}\widetilde{W}_s\right)\,\mathrm{d}V=\int_\Omega \delta\widetilde{W}_s\left[G_s+R_s(\boldsymbol{c}^{\mathrm{bulk}})\right]\,\mathrm{d}V. \tag{27}$$

No analytical infinite-domain field appears in this volume integral.

3 Dipole Tensor in Zr

Point defects interact with the surrounding lattice through their elastic strain fields. At length scales larger than the defect core, this interaction can be represented by an elastic dipole tensor, P_{ij}, which describes the leading-order coupling between a point defect and an externally applied strain field:

$$\Delta E_{\mathrm{def}}=-P_{ij}\epsilon_{ij} \tag{28}$$

P_{ij} acts as a generalized force dipole. Warwick, Ma, and Dudarev computed the elastic dipole tensors of hcp point defects using density functional theory. A point defect was introduced into a periodic supercell, and the atomic positions were relaxed while the cell shape and volume were held fixed. Because the defective crystal was not allowed to relax macroscopically, a residual average stress remained in the supercell. The elastic dipole tensor was then obtained from this residual stress as

$$P_{ij}(N_{\mathrm{def}})=-V\overline{\sigma}_{ij}, \tag{29}$$

where V is the supercell volume and $\overline{\sigma}_{ij}$ is the average residual stress. The negative sign appears because the residual stress is the stress required to constrain the defective crystal at fixed cell shape and volume; the elastic dipole of the defect has the opposite sign. Since the calculation is performed in a finite periodic cell, the defect also interacts with its periodic images. To remove this finite-size effect, the dipole tensor components were computed for several supercell sizes and extrapolated to the isolated-defect limit using

$$P_{ij}(N)=\frac{\alpha_{ij}}{N}+P_{ij}^{\infty}, \tag{30}$$

where N is the number of atoms in the reference supercell, α_{ij} is a fitting constant, and P_{ij}^{∞} is the elastic dipole tensor of an isolated defect. The corresponding relaxation-volume tensor is related to the dipole tensor through

$$\Omega_{ij}=S_{ijkl}P_{kl}^{\infty}, \tag{31}$$

where S_{ijkl} is the elastic compliance tensor. Thus, P_{ij} measures how the defect couples to strain or stress, while Ω_{ij} describes the macroscopic shape and volume change associated with the defect.

3.1 Vacancy

For a vacancy in Zr, the reported elastic dipole tensor is

$$\mathbf{P}^{\mathrm{Vac}}=\begin{pmatrix}-5.156 & 0 & 0\\ 0 & -5.156 & 0\\ 0 & 0 & -8.022\end{pmatrix}\mathrm{eV}. \tag{32}$$

All diagonal components are negative, indicating that the vacancy is a contractive defect. The contraction is anisotropic, and the vacancy produces a larger contractive distortion along the c-axis than in the basal plane. The corresponding relaxation-volume tensor is

$$\boldsymbol{\Omega}^{\mathrm{Vac}}=\begin{pmatrix}-0.085 & 0 & 0\\ 0 & -0.085 & 0\\ 0 & 0 & -0.267\end{pmatrix}\Omega_0 \tag{33}$$

where Ω_0 is the atomic volume of hcp Zr and $\Omega_{\mathrm{rel}}^{\mathrm{Vac}} = \mathrm{Tr}\left(\boldsymbol{\Omega}^{\mathrm{Vac}}\right) = -0.437\Omega_0$. Therefore, the vacancy reduces the crystal volume by approximately 0.44 atomic volumes.

3.2 Interstitial

The lowest-energy self-interstitial configuration listed in the table is the basal-octahedral interstitial, denoted BO. Its elastic dipole tensor is

$$\mathbf{P}^{\mathrm{BO}} = \begin{pmatrix} 16.655 & 0 & 0 \\ 0 & 16.655 & 0 \\ 0 & 0 & 10.344 \end{pmatrix} \mathrm{eV}. \tag{34}$$

All diagonal components are positive, showing that the basal-octahedral self-interstitial is an expansive defect. The expansion is anisotropic meaning that the defect couples more strongly to basal plane strain than to strain along the hcp c-axis. The corresponding relaxation-volume tensor is

$$\boldsymbol{\Omega}^{\mathrm{BO}} = \begin{pmatrix} 0.549 & 0 & 0 \\ 0 & 0.549 & 0 \\ 0 & 0 & -0.008 \end{pmatrix} \Omega_0, \tag{35}$$

where Ω_0 is the atomic volume of hcp Zr. The scalar relaxation volume $\Omega_{\mathrm{rel}}^{\mathrm{BO}} = \mathrm{Tr}\left(\boldsymbol{\Omega}^{\mathrm{BO}}\right) = 1.09\Omega_0$. Thus, the basal-octahedral self-interstitial increases the crystal volume by approximately one atomic volume.

4 Current Numerical Formulation and Consistency Audit

This section records the formulation that is currently implemented in the coupled cluster-dynamics and dislocation-dynamics code and identifies the remaining limitations. In particular, storing the finite-element unknown as W does not by itself transform a physical reaction operator; the required mobile-reaction transformation is therefore included explicitly below.

4.1 Nodal activity and the stress used in the transformation

For each mobile species s, the code uses

$$W_s = H_s c_s, \qquad H_s = \exp\left(\frac{E_s^e}{k_B T}\right). \tag{36}$$

At a finite-element node A, the stress entering E_s^e is the patch-average total stress

$$\boldsymbol{\sigma}_A^{\mathrm{patch}} = \frac{1}{N_A} \sum_{e \in \mathcal{P}_A} \boldsymbol{\sigma}\left(\boldsymbol{x}_e^c\right), \tag{37}$$

where $\mathcal{P}_A$ is the element patch of node A and $\boldsymbol{x}_e^c$ is the centroid of element e. The stress returned by the microstructure container is a sum of all pointwise stress contributions. For the present simulations this is, to leading order,

$$\boldsymbol{\sigma}_A^{\mathrm{patch}} = \boldsymbol{\sigma}_A^{\mathrm{DD,patch}} + \boldsymbol{\sigma}^{\mathrm{ext}}. \tag{38}$$

The currently implemented equilibrium interaction energy is the hydrostatic, scalar-relaxation-volume approximation

$$E_{s,A}^e = -\frac{1}{3}\,\mathrm{tr}\left(\boldsymbol{\sigma}_A^{\mathrm{patch}}\right) n_s \overline{\Omega}_s^{\mathrm{rel}}, \qquad \overline{\Omega}_s^{\mathrm{rel}} = V_s^{\mathrm{rel}} \Omega_0, \tag{39}$$

where n_s is the signed number of point defects in species s and V_s^{rel} is the input `mobileSpeciesRelRelaxVol`. Equation (39) is internally consistent with the current code, but it is not yet the full tensorial coupling

$E_s^e = -\boldsymbol{\sigma} : \boldsymbol{\Omega}_s$. The anisotropic relaxation-volume tensors discussed in the preceding section therefore remain a future extension of the equilibrium interaction model.

Physical concentrations are converted to nodal activities when a `cdMatrix` is read,

$$W_{s,A} = c_{s,A} \exp\left(\frac{E^e_{s,A}}{k_B T}\right), \tag{40}$$

and are converted back on output using the same patch-average stress,

$$c_{s,A} = W_{s,A} \exp\left(-\frac{E^e_{s,A}}{k_B T}\right). \tag{41}$$

Thus, input and output form an exact nodal round trip when the configuration and stress state are unchanged. Immobile-cluster variables are not affected by this transformation. During configuration loading, stress-producing microstructures, including the current dislocation geometry, are initialized before this nodal conversion is performed. The conversion therefore uses the stress of the frame being loaded rather than an empty or previous dislocation configuration.

4.2 External-stress-dependent diffusion tensor

The diffusion tensor used by the finite-element transport operator depends only on the macroscopic external stress. Let $\boldsymbol{Q}_g$ denote the crystal-to-global rotation of grain g. The stress is first rotated into the crystal frame,

$$\boldsymbol{\sigma}^g_{\mathrm{ext}} = \boldsymbol{Q}_g^T \boldsymbol{\sigma}_{\mathrm{ext}} \boldsymbol{Q}_g. \tag{42}$$

For the component ordering

$$\alpha \in \{11, 12, 13, 22, 23, 33\}, \tag{43}$$

the implemented effective migration barrier and local diffusivity are

$$E^{m,\mathrm{eff}}_{s,\alpha} = E^m_{s,\alpha} - \sigma^g_{\mathrm{ext},\alpha} V^m_{s,\alpha} \Omega_0, \tag{44}$$

$$\widetilde{D}^g_{s,\alpha} = D^0_{s,\alpha} \exp\left(-\frac{E^{m,\mathrm{eff}}_{s,\alpha}}{k_B T}\right). \tag{45}$$

Here $V^m_{s,\alpha}$ is read from `mobileSpeciesMigrationRelVolume`. The global tensor is

$$\widetilde{\boldsymbol{D}}^g_s = \boldsymbol{Q}_g \widetilde{\boldsymbol{D}}^{g,\mathrm{local}}_s \boldsymbol{Q}_g^T. \tag{46}$$

No engineering-shear factor is applied to the stress components. Therefore, the migration-volume input must use the same component convention. The present implementation is a component-wise phenomenological parameterization; a jump-resolved saddle-point model would instead evaluate a scalar contraction for every jump direction and then perform the lattice sum in Eq. (4).

The finite-element flux is

$$\boldsymbol{j}^{\mathrm{FEM}}_s = -\frac{1}{\Omega_0} \widetilde{\boldsymbol{D}}^g_s \boldsymbol{\nabla} W^{\mathrm{CD}}_s. \tag{47}$$

Both the flux operator and its species scaling use $\widetilde{\boldsymbol{D}}^g_s(\boldsymbol{\sigma}_{\mathrm{ext}})$. When the external stress changes, the stored stress is updated and the FEM matrix is marked for reassembly. With the current uniform external-load controller, the dislocation network, cluster dynamics, and inclusions have zero average stress, so the stress supplied to the diffusion tensor is the uniform external stress alone. A spatially varying external elastic field is not yet represented by this average-stress path.

4.3 CD–DD superposition and boundary conditions

The implemented decomposition is most accurately written as

$$W_s = W_s^{\mathrm{CD}} + W_s^{\mathrm{DD}}, \tag{48}$$

where W_s^{DD} is the analytical dislocation contribution and W_s^{CD} is the finite-element field. The analytical contribution uses an anisotropic Green-function kernel and includes the effective dislocation bias through a multiplicative $1/Z_s^{\mathrm{EID}}$ factor.

The boundary condition is imposed directly as a Dirichlet condition in activity space. At a boundary node,

$$W_s^{\mathrm{CD}} = W_s^{\mathrm{bnd}} - W_s^{\mathrm{DD}}. \tag{49}$$

Consequently, the sum in Eq. (48) satisfies the prescribed boundary activity. The FEM correction is homogeneous only when irradiation production and reactions are disabled. In the general case, the intended physical equation for the present split is

$$\boldsymbol{\nabla} \cdot \left(\widetilde{\boldsymbol{D}}_s \boldsymbol{\nabla} W_s^{\mathrm{CD}} \right) + G_s + R_s(\boldsymbol{c}^{\mathrm{bulk}}) = 0 \qquad \text{away from the analytical line sources.} \tag{50}$$

The current code realizes Eq. (50) for both the transport-only and reaction-enabled problems. In the latter case, the smooth physical concentration is reconstructed from the nodal activity before the reaction operators are evaluated, as detailed below.

The analytical W^{DD} kernel uses the inverse and determinant of the same external-stress-dependent tensor $\widetilde{\boldsymbol{D}}(\boldsymbol{\sigma}_{\mathrm{ext}})$ as the FEM operator, together with the multiplicative $1/Z_s^{\mathrm{EID}}$ Green-response factor. These tensors are cached per grain and mobile species. The cache is rebuilt whenever the external stress changes, together with the invalidation of the FEM solver matrix. Thus, the analytical and finite-element parts share the same external-stress-dependent diffusion tensor, while the analytical branch additionally carries the effective bias closure defined in Eq. (19). The boundary degrees of freedom are registered once; later stress-triggered matrix reassemblies reuse those constraints and only update their prescribed activity values.

4.4 Reaction terms in the activity formulation

The finite-element unknown and its Newton increment are both stored as W. The first- and second-order reaction operators remain physical concentration operators; consequently, the code reconstructs the smooth bulk concentration before evaluating either the reaction residual or its Newton linearization.

Define the diagonal nodal transformation

$$\boldsymbol{H}_A = \mathrm{diag}\left[\exp\left(\frac{E_{1,A}^e}{k_B T} \right), \ldots, \exp\left(\frac{E_{m,A}^e}{k_B T} \right) \right]. \tag{51}$$

To remain consistent with the reduced model in Eq. (22), the smooth bulk concentration at a quadrature point should be reconstructed from the nodal finite-element activities,

$$\boldsymbol{c}_h^{\mathrm{bulk}}(\boldsymbol{x}) = \sum_A N_A(\boldsymbol{x}) \boldsymbol{H}_A^{-1} \boldsymbol{W}_A^{\mathrm{CD}}. \tag{52}$$

For a physical reaction law

$$R_a(\boldsymbol{c}) = R_{ab}^{(1)} c_b + \frac{1}{2} R_{abg}^{(2)} c_b c_g, \tag{53}$$

the residual must evaluate Eq. (53) using Eq. (52). Its derivative with respect to nodal finite-element activity is

$$\frac{\partial R_a}{\partial W_{s,B}^{\mathrm{CD}}} = \frac{\partial R_a}{\partial c_s} N_B \left(\boldsymbol{H}_B^{-1} \right)_{ss}. \tag{54}$$

The implementation applies the factors $\boldsymbol{H}_B^{-1}$ to the nodal trial shape functions used by both the reaction residual and the reaction Jacobian. The quantity denoted by `R1cd` remains a separate cluster-size scaling and is not used as the c-to-W transformation. The strongly peaked analytical W^{DD} field is deliberately excluded from this volume reaction quadrature, consistently with the inner-reaction approximation in Eq. (21).

Reaction-rate coefficients also contain diffusivities. The first- and second-order coefficient matrices are cached per grain using $\widetilde{\boldsymbol{D}}(\boldsymbol{\sigma}_{\mathrm{ext}})$, and the special one-dimensional sink correction uses the corresponding crystal-frame local diffusivity. These caches are rebuilt together with the analytical diffusion cache whenever the external stress changes. The irradiation source vector is also reassembled because its test-function scaling contains the same stress-dependent diffusivity.

4.5 Output and visualization

The `cdMatrix` stores physical mobile concentrations. DDqt follows the same convention. Before plotting, the nodal CD activity is converted with Eq. (41), and the resulting nodal physical concentration is interpolated with the same quadratic finite-element shape functions used by the solver. The analytical Green response is returned directly as W_s^{DD} and is converted at the visualization point using the total point stress. Therefore, the plotted quantities are

$$c_s^{\mathrm{plot}} = c_s^{\mathrm{CD}} + c_s^{\mathrm{DD}}, \tag{55}$$

and the CD contribution agrees with the output `cdMatrix` at every finite-element node. Before a DDqt frame is initialized, its run ID, physical time, and time step are restored from the corresponding row of the history file. Time-dependent external loads and the resulting diffusion tensor are therefore evaluated at the displayed frame time.

4.6 Implementation status and verification sequence

The present status can be summarized as follows:

Component	Status
Nodal $c \leftrightarrow W$ conversion using patch-average total stress	Implemented for input, equilibrium initialization, output, and CD visualization.
FEM transport with $\widetilde{\boldsymbol{D}}(\boldsymbol{\sigma}_{\mathrm{ext}})$	Implemented, including grain-frame rotation and matrix reassembly after an external-stress change.
Analytical DD kernel with the same external-stress-dependent tensor	Implemented; its inverse and determinant tensors are refreshed together with the FEM operator after an external-stress change.
Diffusion-tensor admissibility	Implemented with a finite-entry and positive-definiteness check before FEM or Green-kernel use.
First- and second-order reactions in the W formulation	Implemented using nodal physical-concentration reconstruction and the Jacobian factors in Eq. (54); the analytical DD field is excluded from bulk reaction quadrature.
External-stress-dependent mobile reaction coefficients	Implemented per grain and refreshed after an external-stress change.
DDqt physical-concentration visualization	Implemented consistently with nodal output, including displayed-frame time synchronization for time-dependent loads.

4.7 Code changes for collaborator review

Table 1 lists the functional code changes in the present revision. Generated build files and the removed historical `GalerkinClimbSolverOld` sources are not included.

Table 1: Code changes implementing the activity formulation and external-stress-dependent diffusion.

Code location	Implemented change	Primary review point
`ClusterDynamicsParameters.h` `ClusterDynamicsParameters.cpp`: parameter initialization and `diffusionTensor()`	Added `mobileSpeciesMigrationRelVolume`; evaluated the stress-dependent migration barriers in the crystal frame using component order $[11, 12, 13, 22, 23, 33]$; rotated the resulting diffusion tensor back to the global frame; rejected non-finite or non-positive-definite tensors before they reach the FEM and Green operators.	Check migration-volume units, barrier sign, component order, and crystal-to-global rotation.
`ClusterDynamicsParameters.h` `ClusterDynamicsParameters.cpp`: activity conversion helpers	Centralized E^e, $c \leftrightarrow W$, and equilibrium conversions. `physicalMobileConcentration()` now evaluates the direct inverse $W/\exp(E^e/k_BT)$ without an exponent clamp.	Check the sign of E^e and the scalar relaxation-volume approximation.
`ClusterDynamicsFEM.h` `ClusterDynamicsFEM.cpp`: transport operator and external stress	Made both `FluxMatrix` and `InvDscaling` use $\widetilde{\boldsymbol{D}}(\boldsymbol{\sigma}_{\mathrm{ext}})$. Added `setExternalStress()`, FEM-matrix invalidation, and per-grain `analyticalInvD`/`analyticalDetD` caches for the analytical Green function. The irradiation source vector is reassembled after an external-stress change because it contains the same inverse-diffusivity test scaling.	Check that only the macroscopic external stress enters diffusion and that a stress change triggers cache refresh and matrix reassembly.
`ClusterDynamicsFEM.h/.cpp`: Dirichlet registration	Separated one-time boundary-degree-of-freedom registration from repeat FEM matrix assembly. External-stress changes now rebuild the matrix without reinserting zero conditions over the current nonzero activity values.	Check repeated external-stress updates with assertions enabled.
`ClusterDynamicsParameters.h/.cpp`: reaction coefficients	Constructed first- and second-order mobile reaction coefficients from the per-grain tensor $\widetilde{\boldsymbol{D}}(\boldsymbol{\sigma}_{\mathrm{ext}})$ rather than a zero-stress diffusion cache. The crystal-frame tensor used by the optional one-dimensional sink correction is evaluated at the same external stress.	Check the scalar averaging of the anisotropic diffusivity and cache refresh after a stress change.
`ClusterDynamicsFEM.h/.cpp` `SecondOrderReaction.h`: mobile reactions	Reconstructed $\boldsymbol{c}_h^{\mathrm{bulk}} = \sum_A N_A \boldsymbol{H}_A^{-1} \boldsymbol{W}_A^{\mathrm{CD}}$ from patch-average total stress and used it in first- and second-order reaction residuals. Applied the same nodal factors to the Newton increment, thereby including the chain-rule factor in the reaction Jacobian.	Check the nodal stress convention, exclusion of W^{DD}, and finite-difference agreement of the reaction Jacobian.
`ClusterDynamicsFEM.cpp`: `initializeConfiguration()`	Interpreted input `cdMatrix` values as physical concentrations and converted them to nodal activities. Applied the same conversion to default equilibrium initialization using the node patch-average total stress.	Check input dimensions and the element-centroid patch average at boundary and interior nodes.

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Table 1 – continued from previous page

Code location	Implemented change	Primary review point
`DefectiveCrystal.cpp`: configuration initialization	Initialized all stress-producing microstructures before Cluster Dynamics so that input $c \to W$ conversion uses the current DD geometry rather than an empty or previous-frame network.	Check first-load and DDqt frame-to-frame input/output round trips.
`ClusterDynamics.h` `ClusterDynamics.cpp`: boundary conditions	Applied the Dirichlet split $W^{\mathrm{CD}} = W^{\mathrm{b}} - W^{\mathrm{DD}}$ at boundary nodes. The prescribed physical boundary concentration is converted to W^{b} using the node patch-average total stress; the analytical W^{DD} returned by the dislocation network is then subtracted.	Check the sign of the superposition, the prescribed boundary concentration, and the point-stress versus patch-average-stress convention for W^{DD} at boundary nodes.
`ClusterDynamics.h` `ClusterDynamics.cpp`: solve and output paths	Updated the external stress before FEM assembly. Added `nodalPhysicalMobileConcentration()` and wrote physical mobile concentrations, rather than activities, to `cdMatrix`.	Check CD-before-DD solve ordering and the nodal $c \to W \to c$ round trip.
`DislocationSegment.h` `DislocationSegment.cpp` and `GalerkinClimbSolver.cpp`	Passed the runtime inverse and determinant of $\widetilde{\boldsymbol{D}}(\boldsymbol{\sigma}_{\mathrm{ext}})$ into the anisotropic analytical kernel and the Galerkin climb stiffness assembly. Returned the Green response directly as W^{DD}, removing the former extra multiplication by H.	Check that the analytical W^{DD} and FEM correction use the same diffusion operator.
`DislocationNetwork.cpp`	Used the runtime diffusion cache in the analytical mobile-activity field, treated the Green response directly as W^{DD}, and converted quadrature-point CD/DD activities to physical concentrations using total point stress on output.	Check the distinction between external stress in $\boldsymbol{D}$ and total stress in $c \leftrightarrow W$.
`DDFieldWidget.h/.cpp` `DefectiveCrystalActor.cpp`	Changed DDqt concentration fields from activity to physical concentration. The CD contribution is reconstructed by interpolating nodal physical concentrations with the quadratic FEM shape functions; other activity contributions are converted at the visualization point using total stress. The displayed frame time is restored from the history file before evaluating external loads.	Check agreement with `cdMatrix` at FEM nodes and visualization-mesh convergence away from nodes.

The mobile reaction residual, reaction Jacobian, and diffusion-dependent reaction coefficients are now consistent with the reduced activity formulation. Immobile-species evolution remains outside the present change. The zero-stress `vClimbRef` is retained because it is used only as a numerical time-step reference and does not enter the climb operator.

The recommended verification and development order is:

1. Verify nodal $c \to W \to c$ round trips at fixed stress to machine precision.
2. Verify that zero external stress reproduces the original diffusion tensors and that prescribed uniaxial and hydrostatic stresses reproduce Eqs. (44)–(46).
3. Check symmetry, positive definiteness, finiteness, and crystal-frame covariance of every diffusion tensor over the intended stress range.

4. Confirm that changing the external stress invalidates and reassembles the FEM matrix.
5. Verify the reaction residual and its analytical Jacobian against finite differences of Eqs. (52) and (54).
6. Verify that all diffusion-dependent reaction coefficients are refreshed from the prescribed external stress.
7. Compare DDqt nodal values against `cdMatrix` output and perform CD-mesh and visualization-mesh convergence studies.
8. Finally, verify tensile/compressive symmetry, far-field behavior, positivity of physical concentration, and convergence of coupled dislocation-climb benchmark problems.

References

[1] Dederichs, P. H., and K. Schroeder. "Anisotropic diffusion in stress fields." Physical Review B 17, no. 6 (1978): 2524.

[2] Da Fonseca, Daphné, Fabien Onimus, Frédéric Mompiou, M-C. Marinica, Edouard de Sonis, Emmanuel Clouet, and Thomas Jourdan. "Numerical investigation of dislocation climb under stress and irradiation." Acta Materialia 242 (2023): 118431.